

Dr. Moin Uddin

Postdoctoral Fellow

King Fahd University of Petroleum and Minerals, Saudi Arabia

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Present Position

October 2025–Present: Postdoctoral Fellow at Interdisciplinary Research Center for Smart Mobility and Logistics, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

Academic Profile

My research interests include Riemannian optimization, variational inclusion problems, and numerical optimization. I am particularly interested in developing efficient algorithms for solving large-scale optimization problems and applying them to machine learning and data-driven decision-making applications.

Research Interests

- Riemannian Optimization
- Numerical Optimization
- Machine Learning
- Variational Inclusion Problems
- Nonlinear Analysis
- Data-Driven Decision-Making

Education

Ph.D. in Mathematics

2019–2024

Aligarh Muslim University, Aligarh, Uttar Pradesh, India

Thesis: Solution Method for Variational and Optimization Problems on Manifolds

Supervisor: Prof. Qamrul Hasan Ansari

M.Sc. in Mathematics | CGPA: 8.92/10

2016–2018

Aligarh Muslim University, Aligarh, Uttar Pradesh, India

B.Sc. (Hons.) in Mathematics | 73.87%

2013–2016

Aligarh Muslim University, Aligarh, Uttar Pradesh, India

Teaching Experience

Assistant Professor

Sep. 2024–Mar. 2025

School of Natural and Applied Sciences

Glocal University, Saharanpur, Uttar Pradesh, India

Teaching Interests

- Advanced Calculus
- Linear Algebra
- Optimization Theory
- Real Analysis
- Metric Spaces
- Functional Analysis

Research Visit

Research Visitor

12 October–11 November 2022

Research Center for Interneural Computing, China Medical University Hospital, Taichung, Taiwan

Department of Applied Mathematics, National Sun Yat-sen University, Kaohsiung, Taiwan

Publications

- [1] Q. H. Ansari, O. P. Ferreira, **Moin Uddin** and M. Alshahrani, Iteration complexity and asymptotic analysis of gradient-type methods with non-monotone line search on Riemannian manifolds, *Journal of Scientific Computing*, 109 (35), 1-36, (2026). doi:10.1007/s10915-026-03456-9
- [2] **Moin Uddin**, M. Alshahrani and Q. H. Ansari, Projection and contraction methods with double inertial steps for variational inclusion problems on Hilbert spaces, *Journal of Computational and Applied Mathematics*, 491, 117993, (2026). doi:10.1016/j.cam.2026.117993
- [3] Q. H. Ansari and **Moin Uddin**, Convergence analysis of incremental quasi-subgradient method on Riemannian manifolds with lower bounded curvature, *Optimization*, (2024) doi:10.1080/02331934.2024.2414782

- [4] Q. H. Ansari and **Moin Uddin**, Modified proximal point algorithms for variational inclusion problems on Hadamard manifolds, *Soft Computing*, 28, 6595-6606, (2024) doi:10.1007/s00500-023-09562-2
- [5] Q. H. Ansari, **Moin Uddin** and J.-C. Yao, Convergence of the Gauss- Newton method for convex composite optimization problems under majorant condition on Riemannian manifolds, *Journal of Complexity*, 80, Article 101788, (2024)
- [6] Q. H. Ansari, **Moin Uddin** and J.-C. Yao, Iterative algorithms for pseudomonotone variational inequalities and fixed point problems on Hadamard manifolds, *Journal of Nonlinear and Convex Analysis*, 24(5), 1043–1063, (2023)
- [7] Q. H. Ansari, Feeroz Babu and **Moin Uddin**, Regularization methods for hierarchical variational inequality problems on Hadamard manifolds, *Arabian Journal of Mathematics*, 12, 309-330, (2023)

Manuscripts Under Review

- [1] Q. H. Ansari, G. C. Bento, O. P. Ferreira, **Moin Uddin** and J.-C. Yao, Logarithmic barrier method on Riemannian manifolds, (2025) **(Under Review at Set-Valued and Variational Analysis)**

Technical Skills

Documentation: LaTeX, Microsoft Office

Programming Tools: MatLab, Julia

Reviewer for Journals

- Journal of Optimization Theory and Applications 2024–Present
- Optimization 2023–Present

Conferences and Workshops

- [1] **Gauss–Newton Method for Convex Composite Optimization Problems on Riemannian Manifolds** (Paper Presentation) 23–25 Jan. 2024
International Halich Congress on Multidisciplinary Scientific Research VII
Istanbul, Turkey (Online)

- [2] **One-Week International Workshop on Technical Writing Using $\text{\LaTeX}$**
(Participant) 30 Mar.–5 Apr. 2024
School of Electronics Engineering (SENSE), VIT-AP University
Amaravati, India
- [3] **Incremental Quasi-Subgradient Method on Riemannian Manifolds with Lower Bounded Curvature** (Paper Presentation) 23–25 Sep. 2023
International Conference on Variational Analysis and Optimization with Applications (ICVAOA)
Aligarh Muslim University, Aligarh, India
- [4] **Iterative Methods for Hierarchical Variational Inequality Problems on Hadamard Manifolds** (Paper Presentation) 1–3 Jul. 2022
International Conference on Dynamical Systems, Control and their Applications (ICDSCA)
IIT Roorkee, India
- [5] **Indo-French Seminar on Optimization, Variational Analysis and Applications** (Participant) 2–4 Feb. 2020
Department of Mathematics, Institute of Science, Banaras Hindu University
Varanasi, India

References

Prof. Suliman Al Homidan

Professor and Dean of Graduate Studies

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Dhahran 31261, Saudi Arabia.

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Dr. Mohammed Alshahrani

Associate Professor

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Dhahran 31261, Saudi Arabia.

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Prof. Orizon Pereira Ferreira

Professor

Institute of Mathematics and Statistics, Federal University of Goiás, Goiânia, Brazil.

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Personal Information

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Nationality: Indian

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